ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3\times10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377\,\Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R,\theta)$ — average power density radiated (W/m^2)
- $F(\theta) = S(\theta)/S_{\text{max}}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D/(4\pi)$ — effective aperture area (m^2)
- A_p — physical cross-section area (m^2)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)

Key shortcuts: *Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{rad} = 73\,\Omega$. 3 dB $\Rightarrow$ linear gain of 2.*

Problem 1 — Hertzian Dipole, Power Density at 5 km — 8 pts

Setup: A 1-m long dipole excited by a 1 MHz current with amplitude $I_0 = 12$ A. Find the average power density radiated at $R = 5$ km, $\theta = 45°$.

DIPOLE TYPE BY LENGTH

Compute first *$\lambda = c/f$, then ratio l/λ*

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$ monopole	1.64	2.15	36.6
$l = \lambda$	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta = S_{\max}\sin^2\theta\ (\text{W/m}^2)$$

$S(R,\theta)$ — avg. pwr dens. at (R,θ) (W/m^2)
l — antenna length (m, the rod itself)
I_0 — peak current in dipole (A)
R — distance from antenna to obs. point (m)
θ — angle from dipole axis (rad/$^\circ$)
$k = 2\pi/\lambda$ — wavenumber (rad/m)
$\eta_0 = 120\pi \approx 377\,\Omega$ — free-space impedance

Max direction *broadside* $\theta_{\max} = \pi/2$

Total radiated power *integrate over 4π*

$$P_{\text{rad}} = \frac{\eta_0\pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2\ (\text{W})$$

Directivity *exact* $D_{\text{Hertz}} = 1.5$ (1.76 dB)

Step 1 — Classify the dipole length. Compute the wavelength to compare with l:

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \quad \Rightarrow \textbf{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ($l/\lambda < 1/50$) means we can use the Hertzian $S(R,\theta)$ formula directly.

Step 2 — Apply the Hertzian power density formula. From HERTZIAN DIPOLE — $S(R,\theta)$:

$$S(R,\theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta$$

Step 3 — Plug in numbers. $\eta_0 = 120\pi$, $k = 2\pi/\lambda = 2\pi/300$ rad/m, $I_0 = 12$ A, $l = 1$ m, $R = 5000$ m, $\theta = \pi/4$:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5\times10^3)^2}\sin^2(\pi/4)$$

Numerator: $(120\pi)(4\pi^2/9\times10^4)(144) = (120\pi)(144)(4\pi^2)/(9\times10^4)$

$$= 120\cdot144\cdot4\cdot\pi^3/(9\times10^4) = 69120\pi^3/(9\times10^4)$$

Denominator: $32\pi^2\cdot2.5\times10^7$. And $\sin^2(\pi/4) = 1/2$.

Working it out cleanly:

$$\boxed{S(5\text{ km}, 45^\circ) = 1.51\times10^{-9}\text{ W/m}^2}$$

Sanity: a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m^2. The $1/R^2$ falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

Problem 2 — Beam Parameters from Normalized Pattern 15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \le \theta \le \pi$ (θ in radians). Find: (a) half-power beamwidth β, (b) pattern solid angle Ω_p, (c) directivity D.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *half-power beamwidth, full angular width at -3 dB*

$$\text{Set } F(\theta) = 0.5, \text{ solve for } \theta \rightarrow \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams, rad or } ^\circ)$$

Gaussian:

$$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$$

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi$$

No ϕ dep.: $\Omega_p = 2\pi \int_0^{\pi} F(\theta)\sin\theta\, d\theta$.

TI-36X Pro: `[2nd][d/dx]` *for* $\int_{\square}^{\square}$, *set RAD mode, multiply result by* 2π.

Narrow-beam Gaussian shortcut ($\sin\theta \approx \theta$, $\rightarrow \infty$):

$$\Omega_p \approx \frac{\pi}{a} = \frac{\pi\,\theta_{1/2}^2}{\ln 2} \quad (F = e^{-a\theta^2})$$

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

Alt. with two orthogonal HPBWs: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.

Step 5 — Gain (if asked) $\xi = $ *rad. eff.*, $0 < \xi \le 1$

$$G = \xi D \qquad G_{\text{dB}} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

(a) Half-power beamwidth β (5 pts). From BEAM PARAMETERS Step 2, the original cheat sheet equations:

$$F(\theta_{1/2}) = 0.5 \Rightarrow \text{ solve for } \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \quad \text{(symmetric beams)}$$

Plug in $F(\theta) = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5)$$

$$\theta_{1/2} = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = 0.186 \text{ rad}$$

That's the half-angle from the peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full half-power beamwidth* is twice that:

$$\boxed{\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ}$$

(b) Pattern solid angle Ω_p (5 pts). From BEAM PARAMETERS Step 3:

$$\Omega_p = \iint_{4\pi} F(\theta)\sin\theta\, d\theta\, d\phi = \int_0^{2\pi}\int_0^{\pi} \exp(-20\theta^2)\sin\theta\, d\theta\, d\phi$$

F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^{\pi} \exp(-20\theta^2) \sin\theta \, d\theta$$

This integral has no closed form; evaluate numerically on the TI-36X Pro.

TI-36X Pro keystrokes (definite integral):

1. Press `[2nd]` $\to$ `[d/dx]` ($\to$ definite-integral template appears, shown as $\int_{\square}^{\square} \square \, d\square$).
2. **Lower limit:** type `0`, then $\to$ arrow to the upper limit.
3. **Upper limit:** press `[π]`, then arrow into the integrand box.
4. **Integrand:** type `e^(-20×x^2)×sin(x)` (use `[e^x]` for the exp, `[x^2]` for the square, `[sin]` for the sine).
5. **Variable:** arrow to the $d\square$ slot, type `x`.
6. Press `[enter]` $\to$ calculator returns ≈ 0.024792.
7. Multiply by 2π: `×[π]×2 [enter]` $\to \approx 0.15578$.

$$\boxed{\Omega_p \approx 0.1558 \text{ sr}}$$

Keep the unrounded calculator value for part (c). Textbook rounds Ω_p to 0.156 first and gets $D = 80.55$; using the full precision gives $D \approx 80.67$. Make sure the calculator is in RADIANS mode (`[mode]` $\to$ RAD) since θ in the exponent is bare — degree mode will give nonsense.

(c) Directivity D (5 pts). From BEAM PARAMETERS Step 4:

$$D = \tfrac{4\pi}{\Omega_p} = \tfrac{4\pi}{0.15578}$$

$$\boxed{D = 80.67}$$

Using textbook's rounded $\Omega_p = 0.156$: $D = 4\pi/0.156 = 80.55$. Use the unrounded value to avoid this rounding error.

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ($\beta \approx 21°$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern 15 pts

Setup: Dipole of length $l = \lambda/4$. Find: (a) directions of maximum radiation, (b) expression for $S_{\max}$, (c) plot of normalized pattern $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right)-\cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

Quarter-wave $l=\lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1-\frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\, S_0$$

Half-wave $l=\lambda/2$ $\pi l/\lambda = \pi/2$, *shortcut form*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\,\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$$

$D = 1.64$, $R_{\text{rad}} = 73\,\Omega$, $\theta_{\max} = \pi/2$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$ monopole	1.64	2.15	36.6
$l=\lambda$	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ *W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.*

General arbitrary-length dipole pattern. From ARBITRARY-LENGTH DIPOLE — $S(\theta)$:

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right)-\cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Plug in $l = \lambda/4$, so $\pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\cos(\pi/4)}{\sin\theta}\right]^2 = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2$$

Let $S_0 = 15 I_0^2/(\pi R^2)$ be the constant prefactor.

(a) Direction of maximum radiation (5 pts). Three-step argument: (i) endpoints via L'Hôpital, (ii) symmetry about $\pi/2$, (iii) sampling to confirm single peak.

Step (i) — endpoints $\theta = 0, \pi$ via L'Hôpital. Inside the squared bracket: $f(\theta) = \cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}$ over $g(\theta) = \sin\theta$.

At $\theta = 0$: $f(0) = \cos(\pi/4) - \sqrt{2}/2 = 0$ and $g(0) = 0 \Rightarrow 0/0$ form. Apply L'Hôpital:

$$f'(\theta) = \frac{\pi}{4}\sin\left(\frac{\pi}{4}\cos\theta\right)\sin\theta, \quad g'(\theta) = \cos\theta$$

$$\lim_{\theta\to 0}\frac{f'(\theta)}{g'(\theta)} = \frac{(\pi/4)\sin(\pi/4)\cdot 0}{1} = 0$$

So $S(0)/S_0 = 0^2 = 0$. Same algebra at $\theta = \pi$ (with $g'(\pi) = -1$) gives $S(\pi) = 0$.

Step (ii) — symmetry about $\theta = \pi/2$. Substitute $\theta \to \pi - \theta$: $\cos(\pi - \theta) = -\cos\theta$, but cos is even so $\cos((\pi/4)\cos(\pi-\theta)) = \cos((\pi/4)\cos\theta)$. And $\sin(\pi - \theta) = \sin\theta$. So $S(\theta) = S(\pi - \theta)$, pattern mirrors about $\pi/2$.

Step (iii) — sample values to confirm single peak (not M-shape). Compute $S(\theta)/S_0$ for a few angles:

θ	30°	60°	90°
S/S_0	0.020	0.063	**0.086**

Monotone rise to 90°, then symmetric fall. Single peak at the center.

$$\boxed{\theta_{\max} = \pi/2 = 90^\circ}$$

(broadside to the dipole axis, same as half-wave dipole.)

(b) $S_{\max}$ (5 pts). Evaluate $S(\pi/2)$:

$$S_{\max} = S(\pi/2) = \tfrac{15I_0^2}{\pi R^2}\left[\tfrac{\cos\left(\frac{\pi}{4}\cdot 0\right)-\frac{\sqrt{2}}{2}}{\sin(\pi/2)}\right]^2 = \tfrac{15I_0^2}{\pi R^2}\left(1-\tfrac{\sqrt{2}}{2}\right)^2$$

Numerically $(1-\sqrt{2}/2)^2 = 0.0858$:

$$\boxed{S_{\max} = 0.0858\,\tfrac{15I_0^2}{\pi R^2}}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts):

$$F(\theta) = \tfrac{1}{0.0858}\left[\tfrac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2$$

Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π. Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Sanity: A $\lambda/4$ dipole has a broadside pattern very close to a Hertzian dipole's $\sin^2\theta$ shape. As $l \to 0$ the bracket approaches $\sin\theta$ and $S \propto \sin^2\theta$ (Hertzian). Half-wave $l = \lambda/2$ has slightly narrower beam than the $\lambda/4$ case.

Problem 4 — Effective Area vs. Physical Cross-section 5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find effective area A_e and compare to physical cross-section A_p.

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi}\ (\mathrm{m}^2)$$

Physical cross section *wire dipole, diameter d*

$$A_p = l\, d = \frac{\lambda}{2}\, d\ \text{(half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2}\ \text{(unitless)}$$

Hertzian special case $D = 1.5$

$$A_e^{\mathrm{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2\ (\mathrm{m}^2)$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$ monopole	1.64	2.15	36.6
$l = \lambda$	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^8} = 3\ \mathrm{m}$$

Step 2 — Directivity for half-wave dipole. From COMMON DIPOLE PARAMETERS: $D = 1.64$.

Step 3 — Effective area. From EFFECTIVE AREA A_e:

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2(1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$\boxed{A_e = 1.17\ \mathrm{m}^2}$$

Step 4 — Physical cross-section. A half-wave dipole has length $l = \lambda/2$. Treat as a rectangle of length l and width d:

$$A_p = l \cdot d = \frac{\lambda}{2} \cdot d = \frac{3}{2}(0.02) = 0.03\ \mathrm{m}^2$$

Step 5 — Ratio.

$$\boxed{\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2}$$

Sanity: The effective area is 39× larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region $\sim\lambda$ around itself.

Problem 5 — Friis Transmission, TV Broadcast 5 pts

Setup: Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t \, F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.
3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{\text{dB}} = 10\log_{10} G_{\text{lin}} \qquad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{5\times10^7} = 6 \text{ m}$$

Step 2 — Antenna gains. TX is half-wave dipole, so $G_t = 1.64$ (from COMMON DIPOLE PARAMETERS).
RX gain: 3 dB → linear via $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$:

$$G_r = 10^{3/10} = 2$$

(3 dB always = ×2, this is the cleanest dB conversion to remember.)

Step 3 — Friis formula. From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$$

Step 4 — Plug in. $R = 30$ km $= 3\times10^4$ m, $P_t = 10^3$ W:

$$P_r = (1.64)(2)\left(\frac{6}{4\pi(3\times10^4)}\right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi\cdot3\times10^4} = \frac{6}{3.77\times10^5} = 1.59\times10^{-5}$$

$$\left(\frac{\lambda}{4\pi R}\right)^2 = 2.53\times10^{-10}$$

$$P_r = (1.64)(2)(2.53\times10^{-10})(10^3) = (3.28)(2.53\times10^{-7})$$

$$\boxed{P_r = 8.3\times10^{-7} \text{ W} = 0.83\ \mu\text{W}}$$

Sanity: 1 kW transmitted, 0.83 μW received at 30 km. That's a 90 dB drop, dominated by the $1/R^2$ free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is −100 dBm $= 10^{-13}$ W), so this link closes comfortably.

Note on the solution PDF: the official key wrote "(1.64)(3)" as a typo, but used $G_r = 2$ in the actual arithmetic (since 3 dB → linear = 2). The final answer 8.3×10^{-7} W matches either way.

Problem 6 — Communication Link with SNR 15 pts (new)

Setup: $G_t = 20$ dB, $G_r = 23$ dB, $R = 20$ km, $P_t = 10$ W, $f = 6$ GHz. All matched. Find: (a) power density at RX antenna, (b) received power, (c) SNR in dB if $T_{\text{sys}} = 1000$ K and $B = 20$ MHz.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t\, F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$. 3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B\, T_{\text{sys}}\, B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

dB ↔ LINEAR

$$G_{\text{dB}} = 10\log_{10} G_{\text{lin}} \qquad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

Setup numbers.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{6\times10^9} = 0.05 \text{ m}$$

$$G_t = 10^{20/10} = 100 \quad G_r = 10^{23/10} = 199.5$$

(a) Power density at RX antenna. TX radiates $G_t P_t$ over a sphere of area $4\pi R^2$:

$$S_r = \frac{G_t P_t}{4\pi R^2} = \frac{(100)(10)}{4\pi(2\times10^4)^2} = \frac{1000}{5.027\times10^9}$$

$$\boxed{S_r = 1.989\times10^{-7} \text{ W/m}^2 \approx 0.199\ \mu\text{W/m}^2}$$

(b) Received power (Friis).

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$$

$$\frac{\lambda}{4\pi R} = \frac{0.05}{4\pi(2\times10^4)} = 1.989\times10^{-7}$$

$$\left(\frac{\lambda}{4\pi R}\right)^2 = 3.957\times10^{-14}$$

$$P_r = (100)(199.5)(3.957\times10^{-14})(10) = 7.894\times10^{-9}$$

$$\boxed{P_r = 7.89 \text{ nW}}$$

Check: equivalent via effective area: $A_{e,r} = G_r\lambda^2/(4\pi) = 199.5(0.0025)/(4\pi) = 0.0397$ m^2, then $P_r = S_r A_{e,r} = (1.989\times10^{-7})(0.0397) = 7.89\times10^{-9}$ W ✓

(c) Signal-to-noise ratio.
Noise floor (from NOISE & SNR):

$$P_N = k_B T_{\text{sys}} B = (1.38\times10^{-23})(1000)(2\times10^7) = 2.76\times10^{-13} \text{ W}$$

$$\text{SNR} = \frac{P_r}{P_N} = \frac{7.89\times10^{-9}}{2.76\times10^{-13}} = 2.86\times10^4$$

$$\boxed{\text{SNR}_{\text{dB}} = 10\log_{10}(2.86\times10^4) = 44.6 \text{ dB}}$$

Sanity: >40 dB SNR is a very clean link. Plenty for digital comm (usually >10 dB suffices).

Problem 7 — Circular Aperture Antenna Scaling 15 pts (new)

Setup: Circular aperture, circular beam $\beta = 3°$ at $f = 20$ GHz. Find (a) D in dB, (b) new D and β if A_p doubled, (c) new D and β if frequency doubled (same A_p).

APERTURE ANTENNAS (horn, dish)

Directivity from aperture area *uniform illumination*

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

Directivity from beamwidths *rectangular beam*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad)}$$

Circular beam ($\beta_{xz} = \beta_{yz} = \beta$): $D \approx 4\pi/\beta^2$.

Half-power beamwidth *uniform rect. illumination*

$$\beta_{xz} \approx 0.88\,\frac{\lambda}{\ell_x} \text{ (rad)} \quad \beta_{yz} \approx 0.88\,\frac{\lambda}{\ell_y}$$

Scaling: $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$. Double area $\Rightarrow$ $D\times 2$, $\beta/\sqrt{2}$. Double f $(\lambda/2)$ $\Rightarrow$ $D\times 4$, $\beta/2$.

dB ↔ LINEAR

$$G_{\mathrm{dB}} = 10\log_{10} G_{\mathrm{lin}} \qquad G_{\mathrm{lin}} = 10^{G_{\mathrm{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

(a) Directivity from beamwidth. For circular beam ($\beta_{xz} = \beta_{yz} = \beta$):

$$D \approx \frac{4\pi}{\beta^2} \quad \text{(use } \beta \text{ in radians)}$$

$$\beta = 3° \cdot \frac{\pi}{180} = 0.05236 \text{ rad}$$

$$D = \frac{4\pi}{(0.05236)^2} = \frac{12.566}{0.002742} = 4584$$

$$\boxed{D = 10\log_{10}(4584) = 36.6 \text{ dB}}$$

(b) Area doubled, $A_p \to 2A_p$. Use $D \propto A_p$ (from $D \approx 4\pi A_p/\lambda^2$):

$$D_{\text{new}} = 2D_{\text{old}} = 9168 \Rightarrow \boxed{D_{\text{new}} = 39.6 \text{ dB}}$$

Since $\beta \propto \lambda/\sqrt{A_p} \propto 1/\sqrt{A_p}$:

$$\boxed{\beta_{\text{new}} = \beta/\sqrt{2} = 3°/1.414 = 2.12°}$$

(c) Frequency doubled ($\lambda \to \lambda/2$), same A_p. Use $D \propto A_p/\lambda^2$:

$$D_{\text{new}} = 4D_{\text{old}} = 18336 \Rightarrow \boxed{D_{\text{new}} = 42.6 \text{ dB}}$$

Since $\beta \propto \lambda$:

$$\boxed{\beta_{\text{new}} = \beta/2 = 1.5°}$$

Sanity: bigger area or shorter wavelength both sharpen the beam and increase directivity. Doubling area gives +3 dB ($\times 2$); doubling frequency gives +6 dB ($\times 4$) because both D and $1/\beta^2$ scale.

Problem 8 — 5-Element Linear Array, $d = 3\lambda/4$ (new)

Setup: Uniform-amplitude broadside linear array, $N = 5$ elements, equal-phase, interelement spacing $d = 3\lambda/4$. Find (a) normalized array factor, (b) HPBW.

LINEAR ANTENNA ARRAY

Setup *N identical elements, spacing d along axis*

element pos $z_i = id,\ i = 0, \dots, N-1;\ A_i = a_i e^{j\psi_i}$

Pattern multiplication

$$S(R,\theta,\phi) = S_e(R,\theta,\phi)\, F_a(\theta)$$

Array factor $F_a(\theta) = |\sum A_i e^{jkid\cos\theta}|^2$

$$F_a(\theta) = \left|\sum_{i=0}^{N-1} A_i\, e^{jkid\cos\theta}\right|^2$$

Uniform $A_i = 1$, **equal phase** *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)},\quad \gamma = kd\cos\theta$$

Normalize: F_a/N^2. *Peak* N^2 *at* $\gamma = 0$ ($\cos\theta = 0$, *broadside*).

HPBW (broadside, uniform) *array length* $L = (N-1)d$

$$\beta \approx 0.88\,\frac{\lambda}{L}\ \text{(rad)}$$

(a) Normalized array factor. From LINEAR ANTENNA ARRAY (uniform, equal-phase):

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)},\quad \gamma = kd\cos\theta$$

With $kd = (2\pi/\lambda)(3\lambda/4) = 3\pi/2$:

$$\gamma = \tfrac{3\pi}{2}\cos\theta,\quad N\gamma/2 = \tfrac{15\pi}{4}\cos\theta,\quad \gamma/2 = \tfrac{3\pi}{4}\cos\theta$$

$$F_a(\theta) = \frac{\sin^2\!\left(\frac{15\pi}{4}\cos\theta\right)}{\sin^2\!\left(\frac{3\pi}{4}\cos\theta\right)}$$

Peak at $\gamma = 0$ ($\cos\theta = 0$, broadside): $F_a^{\max} = N^2 = 25$.

$$\boxed{F_a^{\text{norm}}(\theta) = \frac{1}{25}\cdot\frac{\sin^2((15\pi/4)\cos\theta)}{\sin^2((3\pi/4)\cos\theta)}}$$

(b) Half-power beamwidth. For uniform broadside array, the standard result is:

$$\beta \approx 0.886\,\frac{\lambda}{Nd}\ \text{(rad)}$$

$$\beta = 0.886\cdot\frac{\lambda}{5(3\lambda/4)} = 0.886\cdot\frac{4}{15} = 0.236\ \text{rad}$$

$$\boxed{\beta \approx 13.5^\circ}$$

Check by solving: $F_a^{\text{norm}} = 0.5$ at $\gamma/2 \approx 0.28$ rad, so $\cos\theta = 0.28/(3\pi/4) = 0.119$, $\theta = 83.2^\circ$, offset from broadside (90°) by 6.8°, full HPBW $\approx 13.6^\circ$ ✓.

Sanity: 5 closely-spaced elements give a moderately narrow beam ($\sim 14^\circ$) compared to a single dipole's $\sim 78^\circ$. Adding more elements or increasing spacing further narrows the main lobe (with grating-lobe risk if $d > \lambda$).